

T1: Project Proposal

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 Assign To

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Team Assignment 1: Project Proposal

Before proceeding, recall [what an API is](https://www.mulesoft.com/resources/api/what-is-an-api)  (<https://www.mulesoft.com/resources/api/what-is-an-api>) and recall the examples of services shown in lecture.

Propose your team's idea for a useful service that:

1. maintains some relevant data persistently
2. supports multiple diverse clients
3. provides an application programming interface (API).

Make sure it is truly a service, not an app. Web servers, application servers, content servers, database servers, etc. are not acceptable; your service must "do something" other than just host apps, services or data. If it is deemed that what you described is not a service, you will be allowed to revise after meeting with your project mentor for up to 50% of lost points back.

Maintaining data persistently means that whenever your service terminates and starts up again some arbitrary time later, the data still exists and can be used and modified by later processing. The datastore should be read/write, not append-only. Your service is not required to support crash recovery, in the sense of the machine suddenly losing power or rebooting while the service is running, only intentional termination (shut-down).

Multiple clients means some of the data may be associated with one client and some with another client running the same app. Diverse clients means some of the data may be associated with one client running one app and some of the data may be associated with another client running a different app. There may also be some aggregate or shared data across clients running the same app and even across clients running different unrelated apps. Keep in mind that clients can be other services with their own clients, not just end-user apps.

Your service's API might be a REST API but does not need to be, an RPC-based API is fine too. Keep in mind that you will need to test your API as well as devise some way to demonstrate that your service "works".

Team Formation Reprise:

If your team name, team membership, planned programming language, development platform and/or team github repository has changed since your team formation assignment, please explain. If not, simply state that nothing has changed.

Service Overview:

Write a few paragraphs that provide an overview of the service that your team would like to develop. Explain what it will do and what kind of persistent data it will store and use. You do not need to use any special format like use cases, user stories or OpenAPI.

Client Overview:

Describe at a high level two or more distinct kinds of clients that you envision might use your service. These clients can themselves be either services (with their own clients) or apps (with human end-users). Your team will develop a client app during the second iteration, but it does not need to be one of the potential clients you discuss now.

Testing Overview:

Describe at a high level how you expect to be able to test the API, functionality and internal operation of your service with persistent data but without any clients yet. Your team will expand testing in the second iteration to include your sample client app, but you need to test stand-alone during the first iteration. Think in terms of testing your service as a whole as well as API testing of all individual entry points and unit testing of all individual subroutines.

Submission instructions:

One member of your team should submit a single file (preferably pdf) presenting your preliminary proposal. Your file must contain your team name and the names/unis of all team members. You may submit repeatedly until the deadline. Note that if multiple team members submit, the most recent submission will override all previous submissions by the same or different team members.

If you prefer to have a copy of this document, you may access a PDF version in the Files > Team Assignments > Specs > T1.pdf

Points 100

Submitting a file upload

Due	For	Available from	Until
Oct 18, 2024	Everyone	Oct 8, 2024 at 11:25am	Oct 20, 2024 at 11:59pm

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